

Muon²: Boosting Muon via Adaptive Second-Moment Preconditioning

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IN ONE LINE

One extra step before Newton–Schulz: **lower perplexity with 40% fewer NS steps**, and Muon’s final loss with **up to 23% fewer GPU-hours**.

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PROJECT PAGE

PROBLEM

Fewer Newton–Schulz steps make Muon worse

$$\mathbf{M} \mapsto \mathbf{U}\mathbf{V}^\top$$

Muon orthogonalizes the momentum with 5 Newton–Schulz (NS) steps

3

matrix multiplications per NS step, on the full matrix

+6–20%

validation perplexity with 3 instead of 5 steps

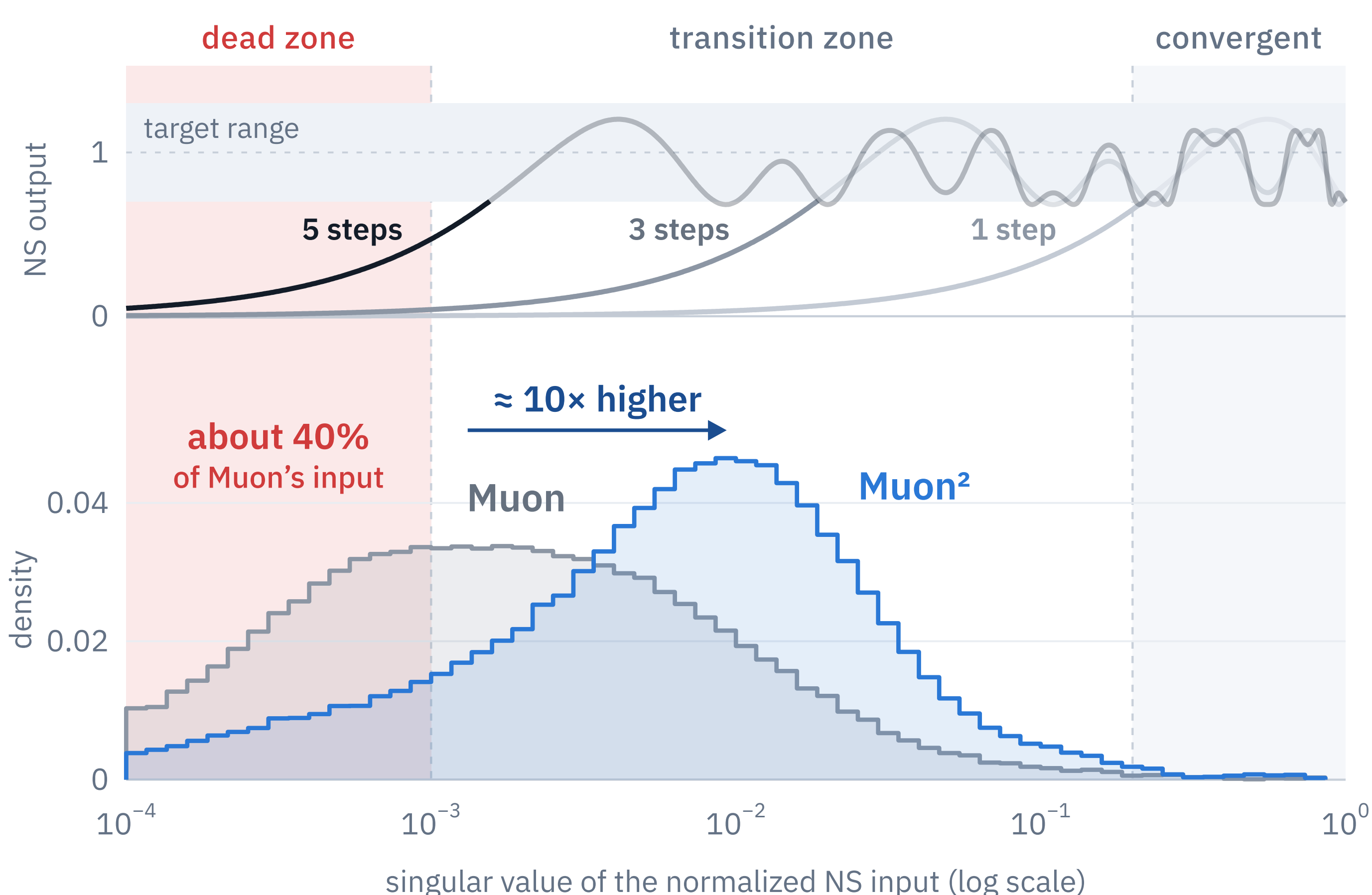
Can we use fewer steps and get a better optimizer?

OBSERVATION

Many singular values are too small for Newton–Schulz

NS output after 1, 3, 5 steps (top) and the singular values entering NS (bottom)

LLaMA-60M, early in training



Small values grow only about 3.4× per step, so below about 10^{-3} five steps are not enough.

METHOD

Muon²: precondition the momentum, then orthogonalize

Muon²momentum $\rightarrow \mathbf{M}_t \oslash (\sqrt{\mathbf{V}_t} + \epsilon) \rightarrow$ Newton–Schulz $\rightarrow$ update

NorMuon, AdaMuon

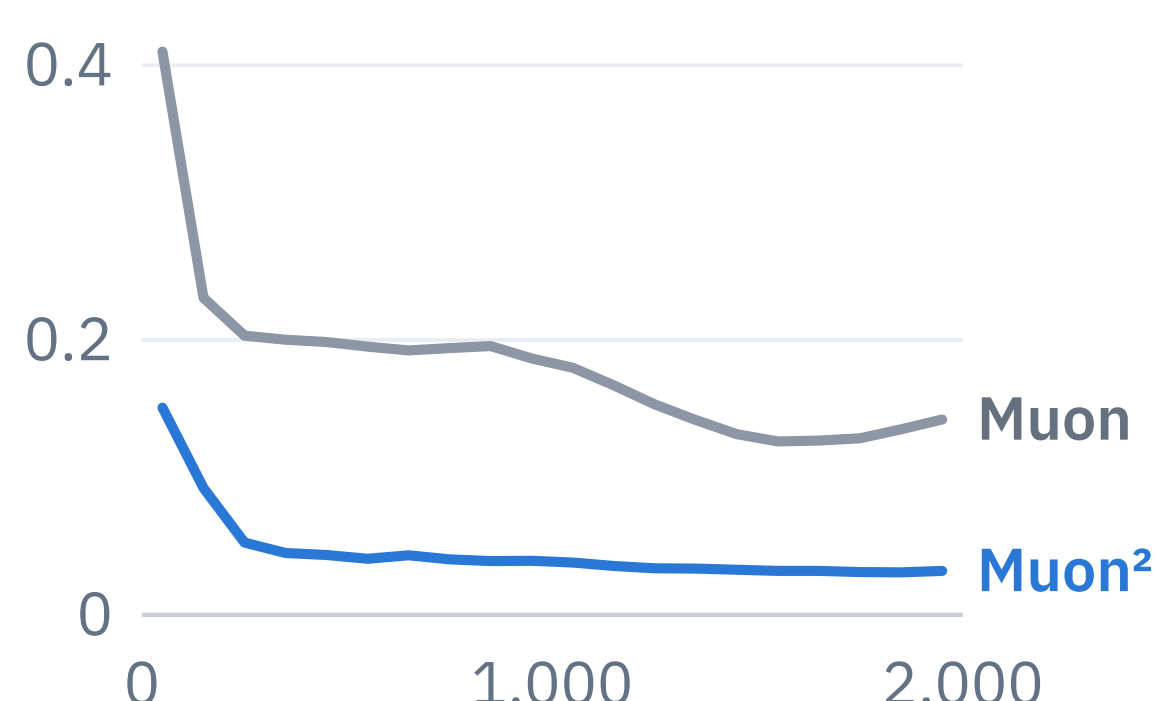
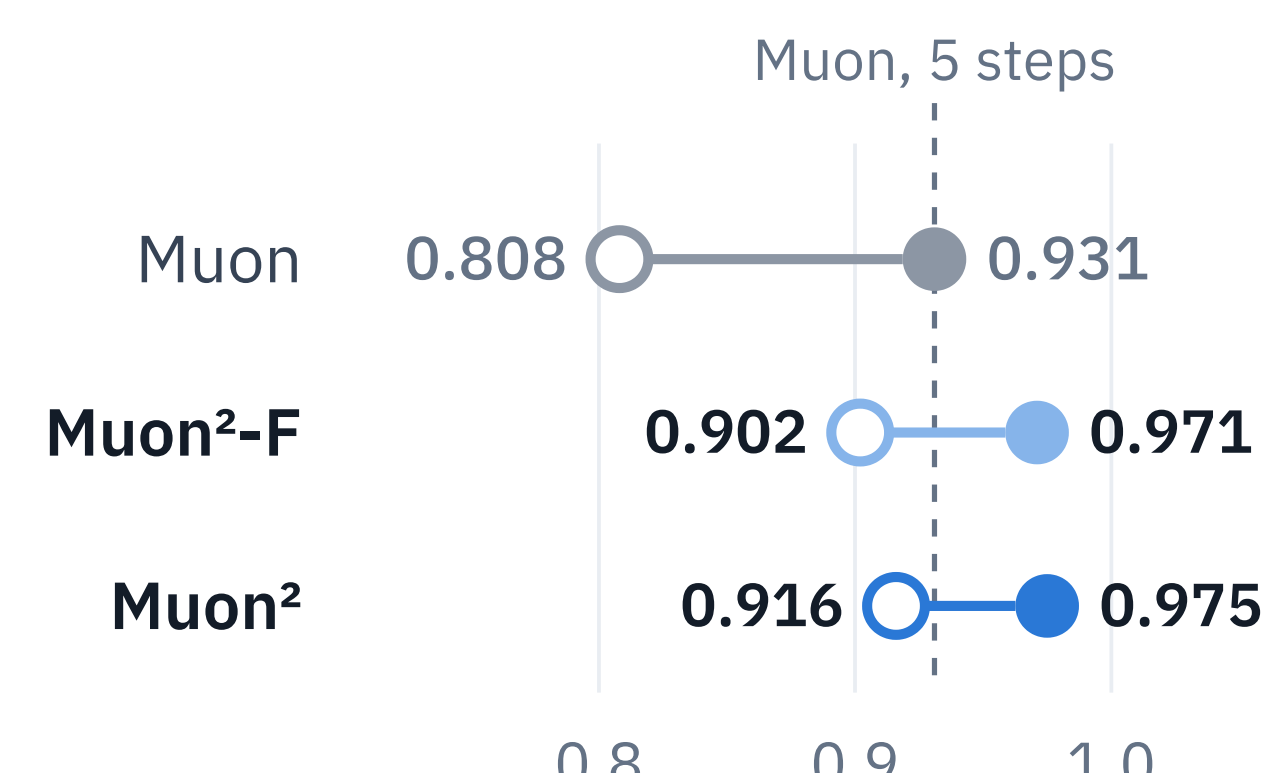
momentum $\rightarrow$ Newton–Schulz $\rightarrow$ rescale by second moment $\rightarrow$ update

$$\mathbf{V}_t = \beta_2 \mathbf{V}_{t-1} + (1 - \beta_2) \mathbf{G}_t \odot \mathbf{G}_t$$
 Adam’s second moment. Everything else is Muon.

The update stays orthogonalized. **Muon²-F** keeps only row and column statistics of $\mathbf{V}$: about the memory of Muon.

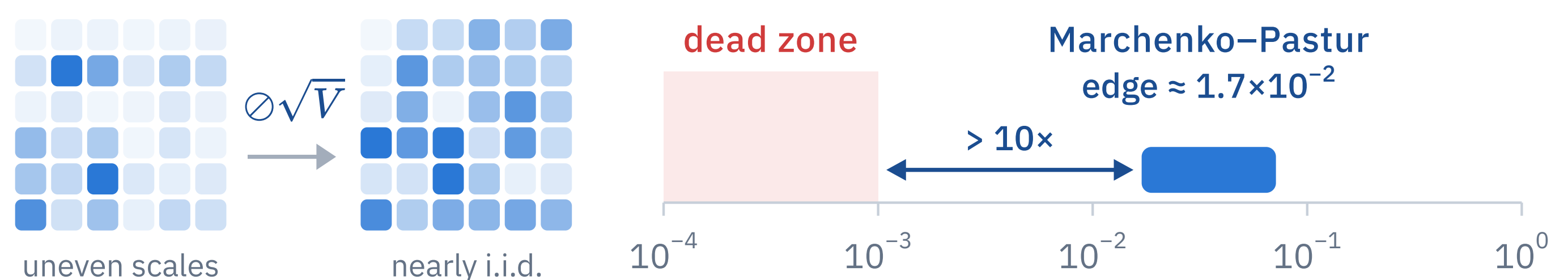
WHY IT WORKS

Out of the dead zone, 3 steps nearly match Muon’s 5

Dead-zone fraction of the NS input ↓
LLaMA-60M, through training**Cosine with the exact $\mathbf{U}\mathbf{V}^\top$ ↑**
open: 3 steps, filled: 5 steps

WHY IT WORKS: THEORY

Dividing by the scale removes the dead zone

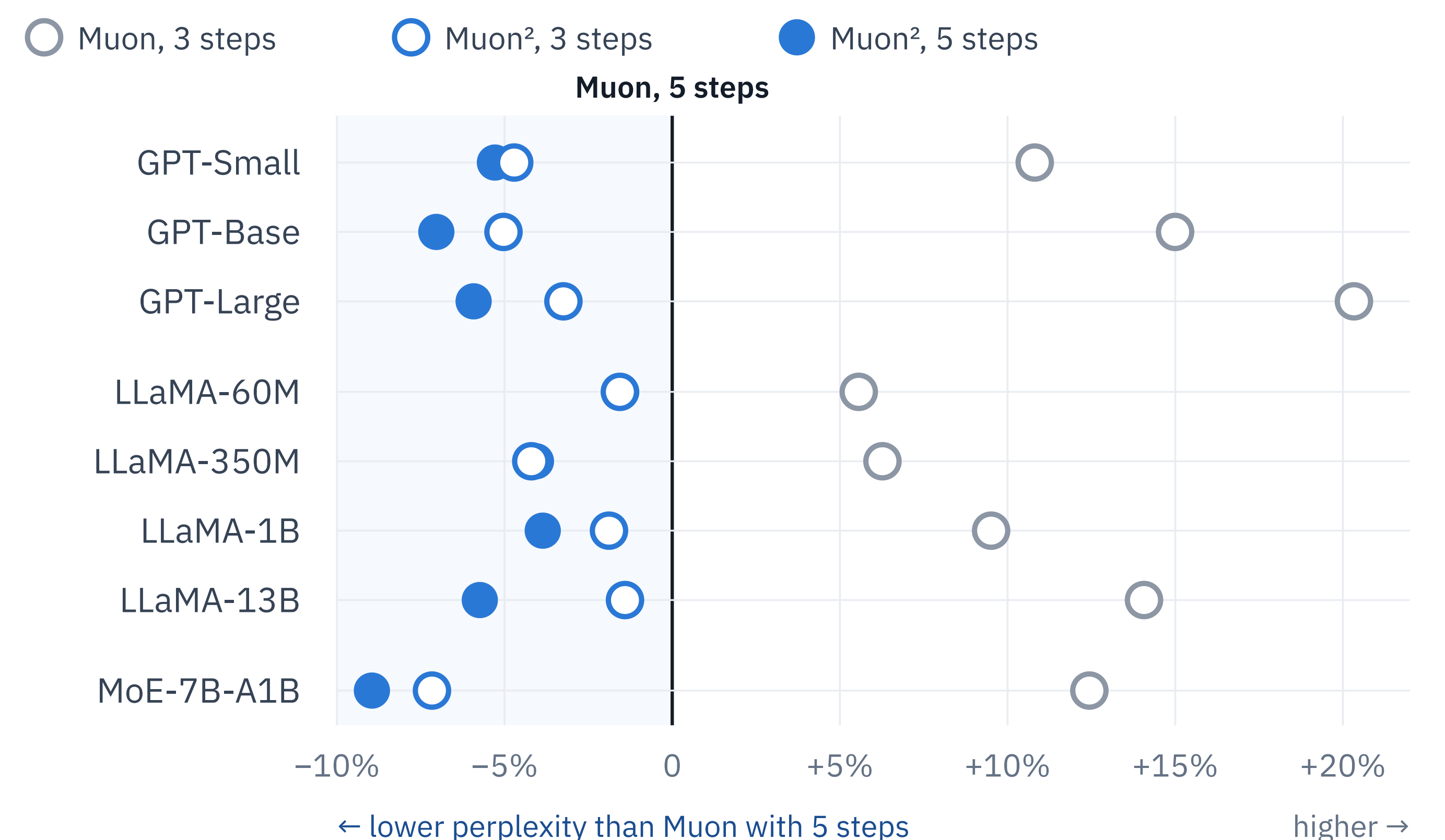
Singular values follow Marchenko–Pastur, away from zeroProposition 1, exact scales, $\lambda = m/n < 1$ · LLaMA-60M MLP layer, $\lambda \approx 0.38$ 

RESULTS

With 3 steps, Muon² beats Muon with 5 on every model

Validation perplexity relative to Muon with 5 steps

GPT on FineWeb · LLaMA 60M to 13B on C4 · MoE 7B, 1B active

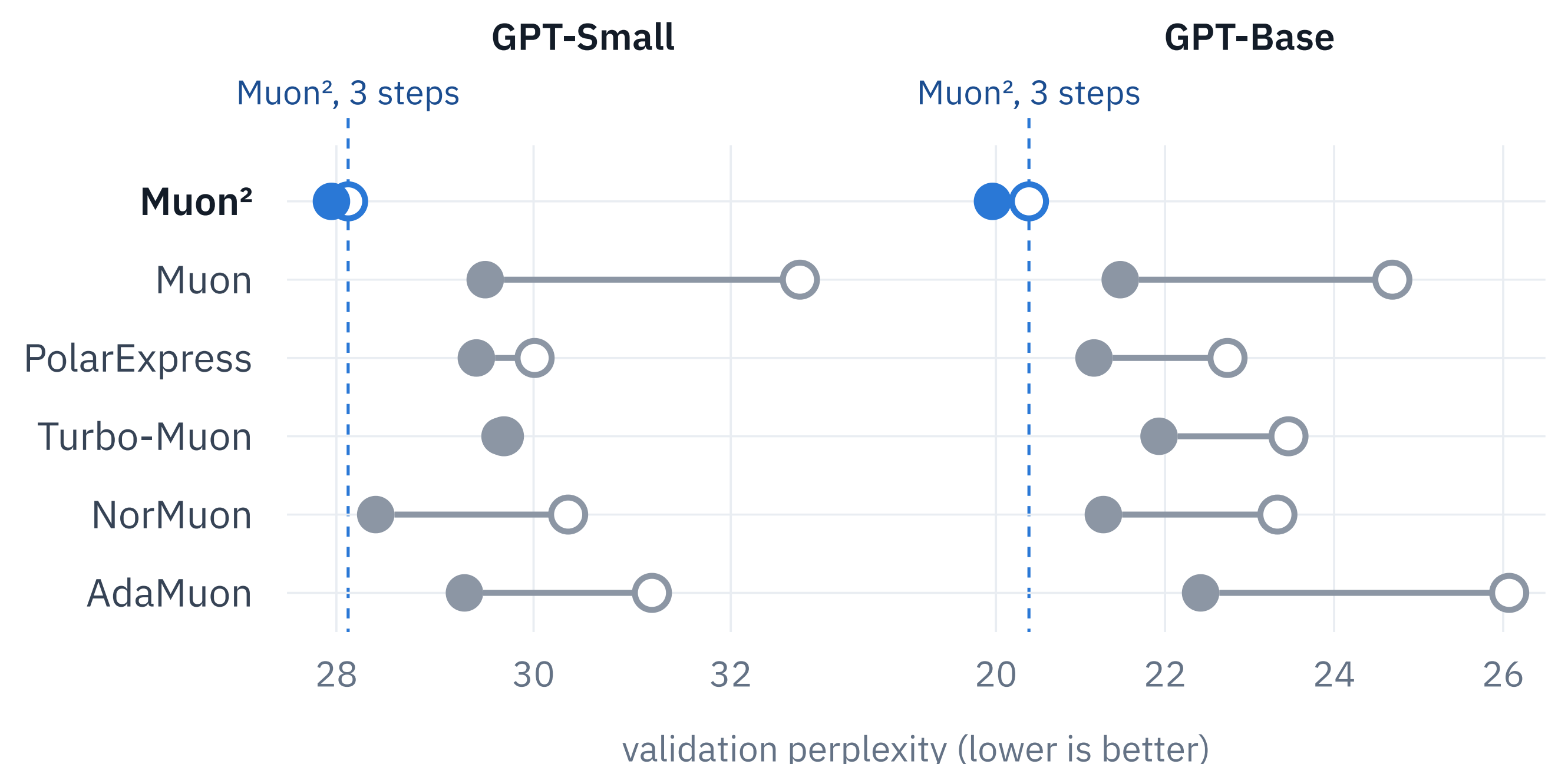


RESULTS

No tuned variant matches Muon², even with more steps

Each method at its best learning rate

open: 3 steps, filled: 5 steps



RESULTS

Muon’s final loss with up to 23% fewer GPU-hours

GPU-hours to reach a loss of 2.36 on LLaMA-1B

2.36 is Muon’s final loss with 5 steps

